

The De-margining Artifact

*does a forecasting model's measured edge over the bookmaker
survive the choice of margin-removal transform?*

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August 17, 2026

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Abstract

Empirical football-forecasting studies routinely claim an edge over the bookmaker. Substantiating such a claim requires converting offered odds into a benchmark probability, which means removing the bookmaker’s margin. At least four transforms are in use for this—proportional normalisation, the power transform, Shin’s insider-trading model, and the constant odds-ratio transform—and papers typically apply one without justification. The four are not equivalent: they differ chiefly in how much margin they attribute to long-shots, which is precisely where a model’s claimed edge tends to concentrate.

We hold a single walk-forward Dixon–Coles model fixed across all four transforms and vary nothing but the benchmark. The comparison covers 176,629 archived matches, 147,177 out-of-sample forecasts, and 91 (league, bookmaker, market) cells drawn from 15 leagues and 14 bookmakers, on 1X2 and Over/Under 2.5 closing prices. Three predictions were pre-registered before the evaluation period was touched: that transform disagreement scales with the book’s margin (P1); that it is larger in the three-outcome market than the two-outcome one (P2); and that there exist cells in which the measured edge changes *sign* between the proportional and Shin benchmarks (P3). P1 and P2 hold: the rank correlation between a book’s margin and the spread across transforms is 0.594, and disagreement is larger in the three-outcome market. P3 does not, in any meaningful sense. Only 1 cell of 91 reverses sign, on returns that are statistically indistinguishable from zero on both sides—because the fixed model achieves no edge anywhere in the sample, losing to the closing line in effectively every cell. The choice of transform demonstrably changes what is measured; whether it can overturn a published conclusion remains open, and this design cannot settle it.

1 Introduction

A bookmaker’s quoted prices do not sum to a probability. For a market with outcomes $i = 1, \dots, n$ and offered decimal odds d_i , the raw implied masses $p_i = 1/d_i$ sum to $B = \sum_i p_i > 1$; the excess $B - 1$ is the overround, the bookmaker’s margin. Any study that claims a forecasting edge must first turn $\{p_i\}$ into a probability vector $\{\pi_i\}$ summing to one, because it is against that vector—not against the quoted prices—that the model’s superiority is asserted.

That conversion is a modelling choice, and the literature has not converged on one. Proportional normalisation is the default. Shin’s model derives a transform from the presence of insider money [12, 13]. The power transform and the constant odds-ratio transform were surveyed against both by Štrumbelj [14], who found their accuracy ordering to depend on market and sport. The four coincide only in the limit $B \rightarrow 1$; away from it they differ in how the removed mass is distributed across outcomes, and they differ most at long odds. This matters because the favourite–longshot bias concentrates market mispricing at exactly those prices [2], and because a model’s largest apparent disagreements with the market are usually found there too.

The question this paper asks is therefore not whether a model can beat the market. It is narrower and, we argue, prior to it:

Does the measured edge of a fixed model over the bookmaker depend materially on which de-margining transform constructs the benchmark?

If the answer is no, the field’s casual treatment of the choice is harmless. If the answer is yes—and in particular if the sign of a measured edge can be flipped by substituting one standard transform for another—then a class of published results is under-determined by its own method section, and reporting the transform becomes a minimum requirement rather than a courtesy.

Our contribution is threefold. First, we isolate the transform as the sole experimental factor: one model, one fit per matchday, one betting rule, four benchmarks. Second, we pre-register the predictions and the frozen model hyper-parameter before the evaluation period is run, so the git history of the pre-registration timestamps the hypotheses ahead of the results (Section 5). Third, we report the full coverage matrix, including every excluded cell, so the reader can see what the study could not measure as well as what it could.

2 The instrument: four de-margining transforms

Each transform maps the raw implied masses $p \in (0, 1)^n$, with book sum $B = \sum_i p_i$, to a probability vector π with $\sum_i \pi_i = 1$.

Proportional. Remove margin uniformly in proportion to implied mass:

$$\pi_i = \frac{p_i}{B}. \tag{1}$$

It fits no free parameter and preserves the ratio of any two implied masses.

Power. Solve for the exponent k satisfying

$$\sum_i p_i^k = 1, \quad \pi_i = p_i^k. \tag{2}$$

Because every $p_i \in (0, 1)$, the sum is strictly decreasing in k , equal to n at $k = 0$ and to B at $k = 1$; a margined book therefore has its unique root at $k > 1$. The transform takes proportionally more margin from longshots, in the direction implied by the favourite–longshot bias.

Shin. Model a proportion z of the money wagered as coming from insiders, and recover [13]

$$\pi_i = \frac{\sqrt{z^2 + 4(1-z)p_i^2/B} - z}{2(1-z)}, \quad (3)$$

with z chosen so that $\sum_i \pi_i = 1$. Note that as $z \rightarrow 0$ this tends to $p_i/\sqrt{B}$, not to p_i/B : Shin does not nest proportional normalisation at its boundary.

Odds-ratio. Impose a constant ratio c between the offered and true odds of every outcome, $p_i/(1-p_i) = c\pi_i/(1-\pi_i)$, which rearranges to

$$\pi_i = \frac{p_i}{c(1-p_i) + p_i}, \quad (4)$$

with c solving $\sum_i \pi_i = 1$. At $c = 1$ the map is the identity and π_i is decreasing in c , so a margined book has its root at $c > 1$ [14].

Three of the four require a root-find. To ensure that no transform is advantaged by a looser solver, all four share a single bisection driver at a common tolerance of 10^{-12} with a common iteration budget, and each transform’s recovered vector is renormalised identically afterwards. The fitted free parameter ($k, z, c; B$ for proportional, which fits nothing) is retained per market so that the effort each transform expended on a given book can be reported.

3 Data

Match results and bookmaker odds come from Football-Data.co.uk [7], which publishes per-league, per-season CSV files containing full-time scores and the closing—and, for recent seasons, opening—prices of a panel of bookmakers.

525 season files were fetched and 514 parsed, yielding 176,629 matches. The eleven unreadable files are recorded rather than silently dropped: seven declare an internal division code that disagrees with their filename, three carry structurally corrupted rows, and one declares the same bookmaker column twice. An unreadable league-season is missing data, and missing data nobody counted is how a coverage table becomes a lie.

Two markets are used: the three-outcome match-odds market (1X2) and the two-outcome Over/Under 2.5 goals market. *Closing* prices only. Opening prices are never used as a benchmark, because a benchmark drawn before the market has absorbed team news measures something other than the market’s final opinion. A match survives into a cell only if its closing book is *complete*—every outcome priced—since a transform applied to a partial book normalises over the wrong support and returns a benchmark that is wrong rather than merely noisy.

The unit of analysis is the (league, bookmaker, market) cell. Cells with fewer than 200 matched fixtures are excluded; the threshold was fixed in the pre-registration rather than chosen after inspection. The full coverage matrix is published as a result, including every excluded cell (Table 1).

4 Method

4.1 Forecasting model

The model is Dixon–Coles [5]: independent Poisson scoring rates with a low-score dependence correction. For a match between home team h and away team a ,

$$\lambda = \alpha_h \beta_a \gamma, \quad \mu = \alpha_a \beta_h, \quad (5)$$

Table 1: Coverage. Complete closing books per bookmaker and market, and the cells that clear the 200-fixture threshold.

Bookmaker	Market	Cells	Matches
Pinnacle	1X2	15	30,769
Bet365	1X2	5	9,838
Bet365	OU25	5	9,838
Bet&Win	1X2	5	9,269
Pinnacle	OU25	5	8,916
William Hill	1X2	5	7,718
VC Bet	1X2	5	7,002
Interwetten	1X2	5	6,243
Betfair Exchange	1X2	5	2,729
Betfair Exchange	OU25	5	2,728
Betfair	1X2	5	1,412
1XBet	1X2	5	1,412
Betfred	1X2	5	1,383
BetVictor	1X2	5	1,371
BetMGM	1X2	5	1,341
Ladbrokes	1X2	3	677
Coral	1X2	3	643

where α is attack strength, β defence weakness, and γ a per-league home advantage. The joint mass of a scoreline (x, y) is

$$P(x, y) = \tau(x, y, \lambda, \mu, \rho) \frac{e^{-\lambda} \lambda^x}{x!} \frac{e^{-\mu} \mu^y}{y!}, \quad (6)$$

with τ the Dixon–Coles adjustment applied to the four scorelines in which both sides score at most one, governed by a single dependence parameter ρ . Fitting is by maximum likelihood over $(\log \alpha, \log \beta, \log \gamma, \rho)$ with a hand-derived analytic gradient; the log parameterisation makes positivity structural rather than a bound, and the analytic gradient is what makes several thousand refits tractable. The gradient is verified against numerical differentiation in the test suite, because a subtly wrong gradient converges silently to the wrong optimum and no downstream check would notice.

The model choice is deliberate and, for this study, incidental. A better model would change the level of every measured edge; the question here is whether the *measurement* moves when only the benchmark is swapped, and that requires the model to be held fixed, not to be good.

4.2 Walk-forward protocol

Each league’s seasons are partitioned in time into three disjoint periods, fixed before any fitting:

Period	Span	Use
Burn-in	first 3 seasons	fitting only; never forecast or scored
Calibration	next 2 seasons	used <i>solely</i> to select ξ
Evaluation	all remaining	the study

Within the scored periods the protocol is: before each matchday, refit on every match that kicked off *strictly* earlier, weighted by $w_m = \exp(-\xi t_m)$ where t_m is the match’s age in days at the cutoff; predict that matchday; roll forward. Burn-in matches remain in the fit window

Table 2: Paired selection of the decay rate on the calibration period. Every candidate is scored on the same 10,567 fixtures.

Half-life (days)	ξ	Paired mean RPS
30	0.023 105	0.222 271
60	0.011 552	0.211 939
90	0.007 702	0.208 691
150	0.004 621	0.206 294
250	0.002 773	0.205 119
400	0.001 733	0.204 889
700	0.000 990	0.205 109
1200	0.000 578	0.205 406
2000	0.000 347	0.205 630
40 000	0.000 017	0.206 014

throughout—excluded from scoring, not from estimation. A matchday whose fit window holds fewer than 300 matches is skipped.

The strict inequality is the single point at which look-ahead leakage could enter, so it is enforced in one place in the code and asserted directly by the test suite.

A fixture is *dropped rather than priced* in three circumstances: a team absent from the fit window has no fitted strength (a promoted side); a fitted rate falls outside $(0.05, 6.0)$ expected goals, which indicates an unidentified parameter rather than a strong team; or ρ is inadmissible for that fixture’s own (λ, μ) . Admissibility depends on the fixture’s own rates, so a ρ valid for every match in the fit window can still be invalid for a pairing not in it, and the fit-time check is structurally unable to catch that.

4.3 The decay parameter, and how weakly it is identified

ξ was selected by minimising mean ranked probability score over the calibration period only, then frozen at

$$\xi = 0.0017328679513999 \quad (\text{half-life } \ln 2/\xi = 400 \text{ days}). \quad (7)$$

Selection was *paired*. Candidates do not price the same fixtures—a fit can fail to converge, or leave a team unidentified—and an unpaired sweep compared means over match sets differing by 43% in size (7,804 to 11,152 forecasts), which is not a valid comparison: an arm that happens to drop harder fixtures scores better for reasons unrelated to forecast quality. That sweep placed its minimum at 700 days, on the edge of its grid. Scoring every candidate on the 10,567 fixtures that *all* candidates forecast moved the optimum to 400 days and made it interior, with the neighbouring candidates worse on either side (Table 2).

Stated plainly: ξ is weakly identified. Half-lives of 400 and 700 days differ by 7.3×10^{-5} in paired RPS, and anything between roughly 250 and 1200 days scores within 6×10^{-4} of the optimum. We do not present 400 days as a precisely determined quantity. What the curve does establish is that the extremes are wrong—a 30-day half-life is far worse (0.222271), and no decay at all is worse than moderate decay (0.206014 against 0.204889)—so exponential decay earns its place without its exact rate mattering much. Because one model is held fixed across all four transforms, a weakly identified ξ shifts every arm together and cannot generate a difference between them.

4.4 Metrics

Ranked probability score. Forecast accuracy is scored with the ranked probability score [6], which respects the ordering of 1X2 outcomes and is the appropriate proper score for this

market [3, 8]. For a forecast π over n ordered categories and realised category j ,

$$\text{RPS} = \frac{1}{n-1} \sum_{k=1}^{n-1} \left(\sum_{i \leq k} \pi_i - \mathbf{1}[j \leq k] \right)^2. \quad (8)$$

Both the model and each de-margined benchmark are scored on the same fixtures, so the comparison is paired by construction.

Economic edge. Prediction P3 is an economic statement, so it is measured economically. The betting rule is deliberately minimal: back an outcome when the model’s probability exceeds the de-margined market probability for that outcome, stake one unit, and settle at the *offered* price. Only the benchmark varies between arms; varying the strategy as well would make any difference in measured edge unattributable. Return on investment is total profit over total turnover, and is reported as undefined—not zero—when a transform selected no bets, since “no bets placed” and “bets placed that broke even” are different findings.

Score decomposition. Where the source of a difference in accuracy is at issue, the Brier score is decomposed following Murphy [9]. We use the exact five-term form

$$\text{BS} = \text{REL} - \text{RES} + \text{UNC} + \text{WBV} - 2 \text{COV}, \quad (9)$$

whose last two terms—within-bin forecast variance and within-bin forecast–outcome covariance—are identically zero only for single-valued bins. On binned continuous forecasts they are not; see Appendix A.

4.5 Inference

Three tools, each chosen because the naive alternative is invalid on this data rather than merely less powerful.

Forecast comparisons use the Diebold–Mariano test [4] with Newey–West HAC standard errors [10] at the standard truncation lag $\lfloor 4(n/100)^{2/9} \rfloor$. Forecasts produced by overlapping fit windows are serially correlated, and treating per-match score differences as independent would understate the variance and overstate significance.

Interval estimates for ROI use a stationary block bootstrap [11] with geometric blocks of mean length 20 and a fixed seed. Betting returns are heavy-tailed and clustered by matchday; an i.i.d. bootstrap would resample away exactly the dependence that makes the true interval wide.

The study runs a family of tests across leagues, bookmakers, markets and transform pairs, so p -values are controlled for false discovery rate by the Benjamini–Hochberg procedure [1] at $q = 0.10$ across that family.

5 Pre-registration

The predictions, the protocol, the exclusion threshold and the frozen ξ were committed to the repository before the evaluation period was run. The git timestamp on `paper/preregistration.md` is the claim; if that file’s history shows modification after the first evaluation run, every prediction below should be treated as post-hoc. The evaluation script parses ξ out of the committed pre-registration and refuses to start unless it agrees with the calibration output to 10^{-9} —a rounded value in the document would be a *different* decay rate from the one selected, and registering one number while running another would defeat the point of registering it.

ID	Prediction	Falsified by
P1	Disagreement between transforms scales with the book’s margin: negligible on a $\approx 102\%$ book, material on a $\approx 107\%$ book	Transform spread on high-margin cells not exceeding that on low-margin cells
P2	Disagreement is larger for three-outcome 1X2 than for two-outcome Over/Under 2.5	OU25 spread matching or exceeding 1X2 spread on the same cells
P3	There exist (league, bookmaker) cells where the measured edge <i>changes sign</i> between proportional and Shin	No cell showing a sign reversal in ROI between those two transforms

P3 is the headline: sign reversal, not merely magnitude change, is what would invalidate a published conclusion. We committed in advance to reporting all three predictions with effect sizes and confidence intervals regardless of outcome, to dropping or reframing none of them after seeing results, and to writing a null on P3 with P1 and P2 holding as a complete paper.

6 Results

The evaluation produced 147,177 out-of-sample forecasts over 91 cells spanning 15 leagues and 14 bookmakers.

6.1 P1: disagreement scales with the book’s margin

P1 is supported. Across the 14 bookmakers the rank correlation between a book’s mean sum and the spread of the four transforms within a cell is 0.594. The pattern in Table 3 is monotone at the extremes: the two sharpest books, Betfair Exchange at a mean sum of 1.008 and Pinnacle at 1.031, both show a mean RPS spread of 0.0001, while books between 1.06 and 1.09 show 0.0004 to 0.0009. The transforms agree almost exactly where there is little margin to remove and diverge by four to nine times as much where there is more of it, which is what the limiting argument predicts.

The stated falsification condition was that the spread on Bet365 should fail to exceed the spread on Pinnacle. It does exceed it, on both measures (0.0110 against 0.0086 in ROI). P1 stands.

Two cautions. The ROI spread column is noisier than the RPS column and does not order as cleanly, because ROI is a function of how many bets a cell generated and the smallest cells here rest on three league-seasons. And the mechanism is not in doubt—the transforms provably coincide as the book sum approaches one—so what this measures is the size of the effect at realistic margins, not whether it exists.

6.2 P2: disagreement is larger in the three-outcome market

P2 is supported, weakly. On the cells where the same bookmaker prices both markets, the mean ROI spread is 0.0111 for 1X2 against 0.0075 for Over/Under 2.5, and the 1X2 spread is the larger in 60% of pairs on ROI and 80% on RPS.

The evidence is thin and should be read as such. Only three bookmakers in the archive carry closing prices for both markets, which leaves fifteen paired cells. The direction matches the prediction and the mechanism is clear—the transforms differ chiefly in how they treat a longshot, and a near-even two-outcome book does not have one—but fifteen pairs cannot distinguish a moderate effect from a small one.

Table 3: Transform disagreement against book margin. Spread is the range of a quantity across the four transforms within a cell.

Bookmaker	Cells	Mean book sum	Mean ROI spread	Mean RPS spread
Betfair Exchange	10	1.0080	0.0085	0.0001
Pinnacle	20	1.0309	0.0086	0.0001
1XBet	5	1.0440	0.0317	0.0004
Coral	3	1.0575	0.0489	0.0005
Interwetten	5	1.0624	0.0291	0.0005
Bet365	10	1.0625	0.0110	0.0003
Ladbrokes	3	1.0638	0.0632	0.0004
VC Bet	5	1.0675	0.0293	0.0005
Bet&Win	5	1.0679	0.0236	0.0005
BetMGM	5	1.0744	0.0515	0.0004
BetVictor	5	1.0760	0.0536	0.0004
William Hill	5	1.0761	0.0202	0.0004
Betfred	5	1.0830	0.0520	0.0005
Betfair	5	1.0898	0.0294	0.0009

Table 4: Transform disagreement by market, on cells where both markets are available from the same bookmaker.

Quantity	Pairs	1X2	OU25	Difference	Share 1X2 larger
ROI spread	15	0.0111	0.0075	0.0036	0.6000
RPS spread	15	0.0002	0.0001	0.0001	0.8000

6.3 P3: sign reversal of the measured edge

P3 is met on its letter and fails on its substance, and the failure is the more informative of the two.

Exactly 1 cell of 91 (1.1%) reverses the sign of measured ROI between proportional normalisation and Shin’s method. On the prediction as written—*there exist* cells where the edge changes sign—that is a positive result. Table 5 shows why it should not be read as one.

The cell is B1/Coral/1X2 on 214 matches, barely above the 200-fixture threshold fixed in the pre-registration. Proportional normalisation puts the ROI at +0.0020 and Shin at −0.0004. Both bootstrap intervals span approximately ± 0.10 . These are not two different conclusions about an edge; they are the same conclusion—no measurable edge—with noise falling on opposite sides of zero. A reversal between two quantities that are each indistinguishable from zero carries no information about whether the choice of transform can overturn a finding.

The deeper problem is that the premise of P3 does not hold in this data, for reasons set out in the next section: the model has no edge anywhere, so there is no edge whose sign could be reversed. P3 was constructed on the assumption that a competently implemented Dixon–Coles model would show positive measured edge in at least some cells, against at least the softer books. It does not. The prediction was therefore not so much falsified as rendered untestable by this design, which is a limitation of the study rather than a finding about the transforms.

6.4 Forecast accuracy against the benchmark

The model is beaten by the market comprehensively and almost uniformly. Table 6 gives the picture: mean model RPS of 0.2053 against benchmark RPS between 0.1987 and 0.1990, a gap

Table 5: Cells in which the sign of measured ROI differs across transforms, with bootstrap intervals.

League	Bookmaker	Market	Matches	ROI (prop.)	ROI (Shin)	Spread
B1	Coral	1X2	214	0.0020	−0.0004	0.0184

Table 6: Model versus benchmark RPS by transform, with Diebold–Mariano statistics and Benjamini–Hochberg adjusted p -values.

Transform	Cells	Model RPS	Benchmark RPS	Difference	Share favouring model	Share $q < 0.10$
Proportional	91	0.2053	0.1990	0.0063	0.0110	0.7582
Power	91	0.2053	0.1987	0.0065	0.0000	0.7473
Shin	91	0.2053	0.1988	0.0065	0.0000	0.7582
Odds-ratio	91	0.2053	0.1988	0.0065	0.0000	0.7582

of roughly 0.0065 in the market’s favour under every transform. The model achieves a lower RPS than the de-margined benchmark in *no* cell under power, Shin or the odds ratio, and in one cell of 91 under proportional normalisation. Around three-quarters of cells reject equal predictive accuracy at $q < 0.10$ after the Benjamini–Hochberg correction, in every case in the market’s favour.

Economic returns agree. Mean ROI across cells ranges from −7.9% under proportional normalisation to −9.4% under the power method, and between 9% and 11% of cells show positive ROI at all. Exactly one cell of 91 has a bootstrap interval lying entirely above zero, which is fewer than the four or five that would be expected by chance across this many cells.

None of this is surprising and none of it indicates a defect. Dixon–Coles is a 1997 model with two parameters per team, no player information, no market data and no in-play adjustment, evaluated against closing prices that aggregate everything the betting market learned before kickoff. That it loses is the expected outcome. It is reported here because it determines what P3 can mean, and because a study that measured an edge without checking whether one existed would be reporting an artefact of its own arithmetic.

7 Discussion

The methodological claim survives and the headline claim does not.

That the four transforms diverge, and that they diverge in proportion to the margin being removed and more sharply where a longshot is present, is established here on 91 cells across 15 leagues and 14 bookmakers. A paper that de-margins a soft book by proportional normalisation and one that uses Shin’s method are not measuring the same benchmark, and the gap between them is largest exactly where a model’s claimed edge tends to concentrate. Reporting which transform was used, and ideally reporting a result under more than one, costs nothing and is not current practice.

What this study cannot show is that the choice flips a conclusion, because our model never reaches a conclusion worth flipping. The design held the model fixed so that any difference between arms would be attributable to the transform alone—which was the right control—but it assumed the fixed model would land near the market. It does not come close. Against that backdrop the single sign reversal we observe is noise around zero, and would be noise around zero whichever transform produced it.

The honest reading is that this work establishes the mechanism and bounds its size, while leaving the question it set out to answer open. Settling it needs a model that actually competes

with closing prices in some segment—a stronger forecaster, a softer or slower market, or an early line rather than a close—so that there is a positive edge on which the transforms can be made to disagree. That is a different study, and the apparatus built here would run it unchanged.

8 Limitations

The model is fixed by choice. This study says nothing about whether a better model would beat the market; it asks only whether the measurement of any model’s edge depends on the transform. A model with a different error profile would concentrate its disagreements with the market at different prices, and the size—though not the existence—of the effect reported here is specific to this model.

Only two markets and only closing prices are used. Nothing here generalises to in-play markets, Asian handicaps, or opening lines.

Backtested returns are not achievable returns. There are no stake limits, no line movement between the moment a price is observed and the moment a bet could be struck, no commission, and no account restrictions. The ROI figures exist to be compared *across transforms*, which is what the design controls; their absolute level is not a claim about what a bettor would have earned.

The decay parameter is weakly identified (Section 4.3). This does not threaten the comparison, which holds it fixed across arms, but it does mean the model should not be read as tuned.

The most consequential limitation is that the fixed model has no measurable edge anywhere in the sample. P3 was framed on the assumption that it would show positive edge in at least some cells against at least the softer books; it does not, in any cell, under any transform. Everything reported about P3 is therefore a statement about behaviour near zero rather than about whether the transform choice can overturn a real finding. P1 and P2 are unaffected, since both concern disagreement between benchmarks and neither requires the model to be any good.

The calibration period is early. A fixed three-season burn-in and two-season calibration on a 33-season archive places the selection of ξ in 1996–97 for the longest-running leagues, up to 28 years before the evaluation period ends. The split was fixed in the design specification before any data was examined and was deliberately not revised once that consequence became visible.

9 Reproducibility

All code, the pre-registration, and the build for this paper are in the repository at <https://github.com/Bardakor/The-De-margining-Artifact>. The pre-registration cites the repository’s former name, which now redirects; that file is left byte-for-byte as committed, because its value rests on not having been edited after the fact. The study runs in three stages—**coverage**, **calibrate**, **evaluate**—in that order, with the pre-registration commit required between the second and the third. Every stochastic step takes an explicit seed and no global random state is touched. The tables in Section 6 are generated mechanically from the evaluation output rather than transcribed.

A The five-term Murphy decomposition

Murphy’s vector partition [9] is usually quoted in three terms, $BS = REL - RES + UNC$, which is exact only when every bin holds a single distinct forecast value. Binning continuous forecasts leaves a residual. Expanding $\sum_i (p_i - o_i)^2$ within a bin around the bin means $\bar{p}, \bar{o}$ gives

$$(p_i - o_i) = (p_i - \bar{p}) + (\bar{p} - \bar{o}) + (\bar{o} - o_i), \quad (10)$$

and squaring leaves three squared terms plus one surviving cross term, $-2(p_i - \bar{p})(o_i - \bar{o})$. Summed, that is minus twice the within-bin *covariance* between forecast and outcome. It vanishes only when, inside every bin, higher forecasts carry no information about which events occurred.

The four-term form that stops at within-bin variance is therefore one term short. It is easy to miss because worked examples with one or two points per bin satisfy the zero-covariance condition by accident. On real continuous forecasts it does not vanish: measured on 4,000 simulated forecasts the covariance term accounted for 1.2% of the Brier score, which the four-term form silently misattributes to reliability and resolution. Including it drops the reconstruction residual from order 10^{-3} to machine zero. All five terms are reported, and the residual is computed explicitly so the identity can be checked rather than trusted.

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